LiDAR-Based Mapping and the Parameterization of a Hydrological Model

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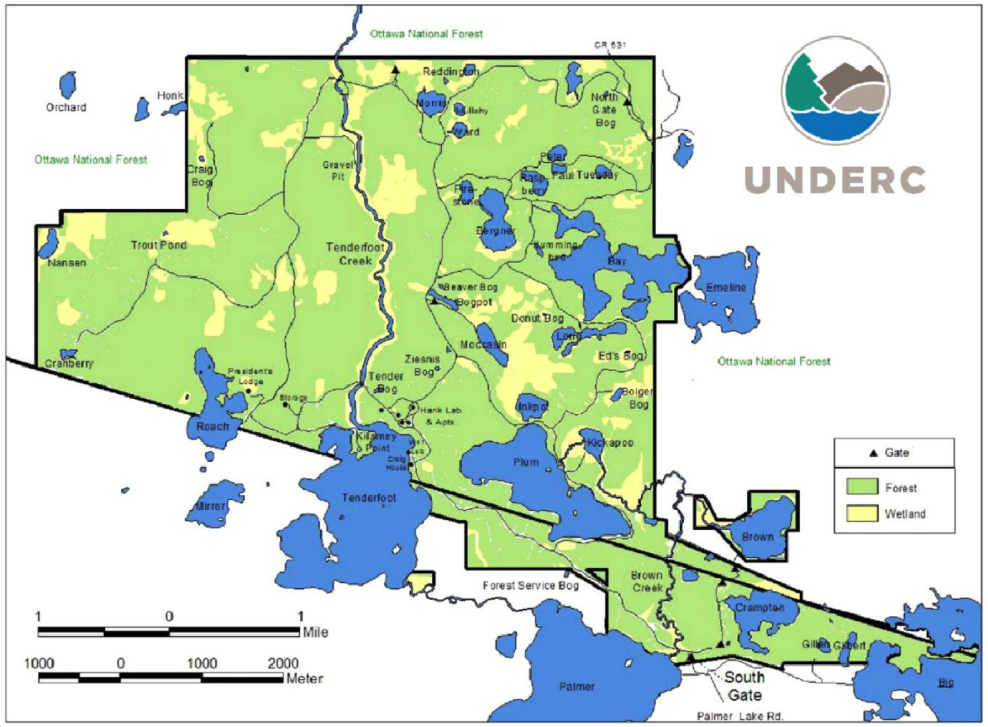
Why this work matters

- UNDERC is in the Northern Highlands Lake District in northern Wisconsin and has one of the highest densities of lakes in the world!
- A hydrological model gives insight to water and nutrient flows



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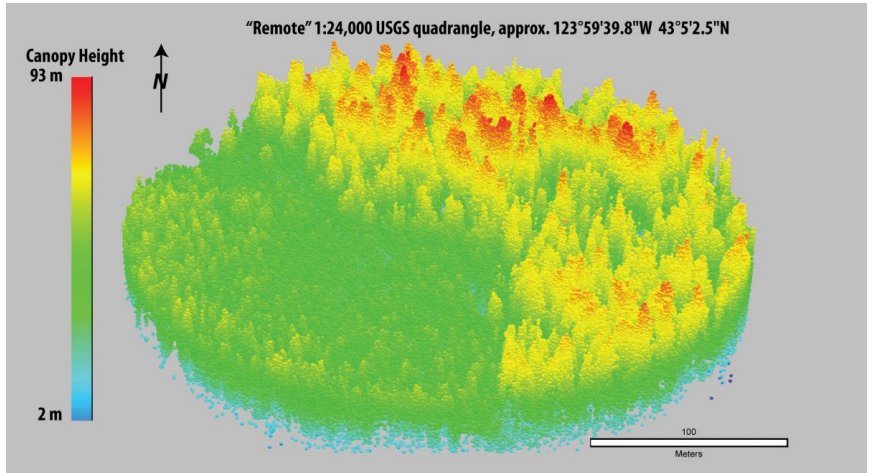
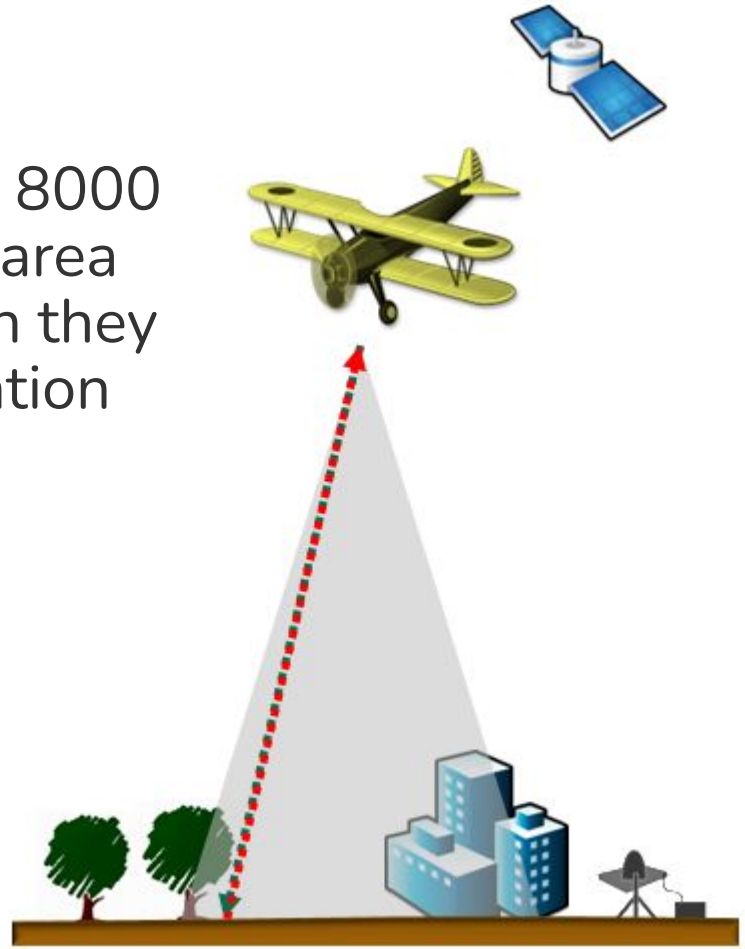
LiDAR Data - light detection and ranging

- Collected by NEON at a 1x1 meter resolution over the 8000 acres UNDERC property and more of the surrounding area
- a plane transmits light pulses to the ground, and when they reflect back, their travel time is used to calculate elevation





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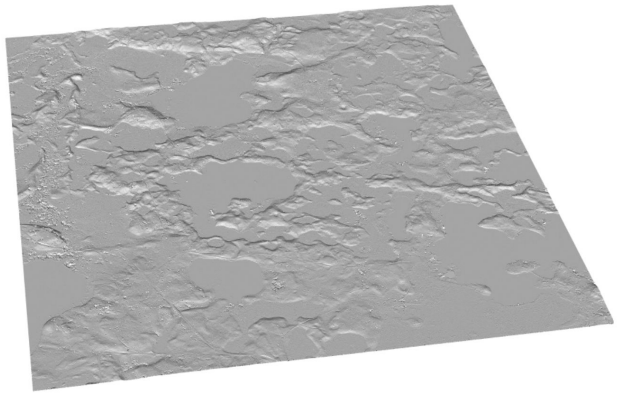
DTM - Digital Terrain Model



nine square km
of the final dtm

The steps to get to a raster:

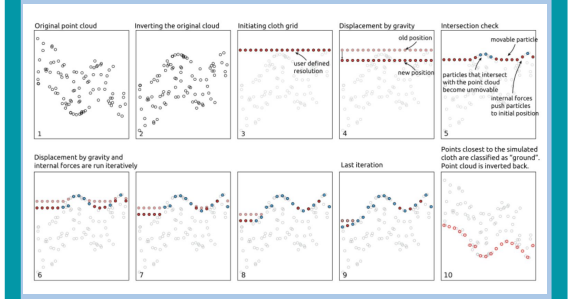
Point Cloud

Issue: “birds” in the sky, points below the surface creating divots in the final surface

Solution: filtering out upper and lower percentiles per square

Ground Point Classification

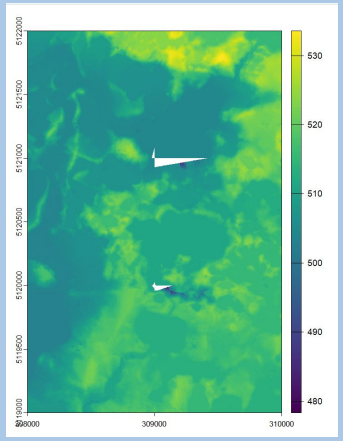
the LiDAR package includes many methods with similar results



Interpolating Points

Using kriging (method that finds mean elv. between points)

Issue! Sheen on lakes (missing points at the corner of lake cells)



Solution: using LAScatalog functions that include a buffer from surrounding cells

Finally, plot and write the raster to a folder

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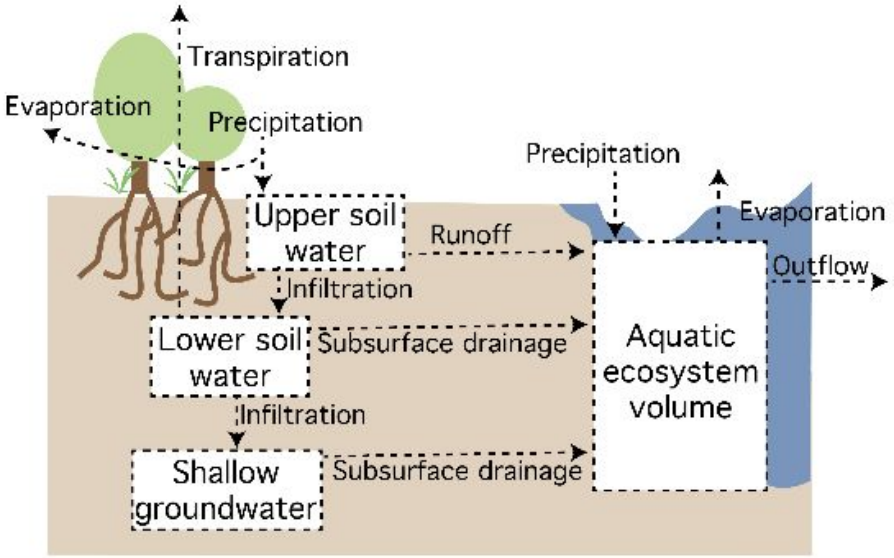
LWB model overview



Our state variables: Lake volumes as a vector [V1, … , Vn] and the heights of upper/middle/lower soil layers

For a given lake…

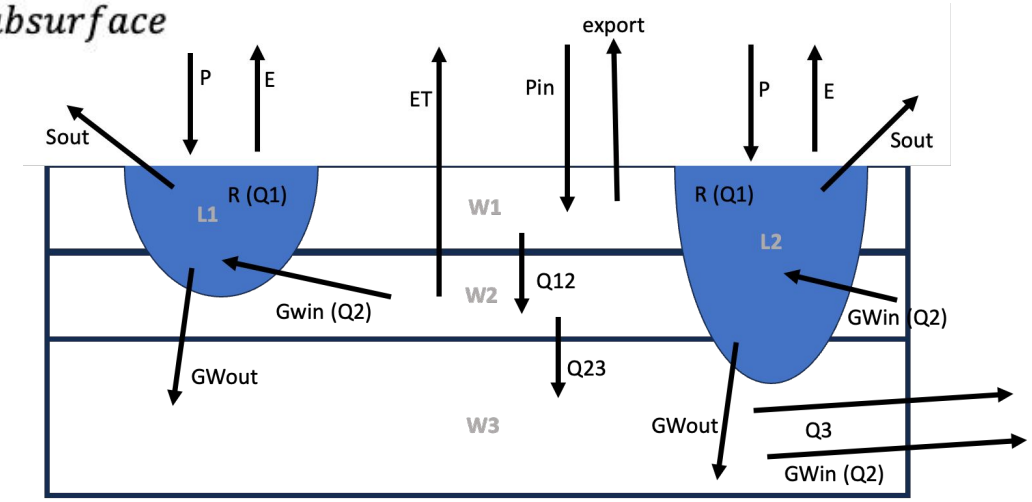
$$\frac{dV}{dt} = P + GW_{in} + R - GW_{out} - E + S_{flowout} + I_{subsurface}$$

Across the property…

$$\frac{dW_1}{dt} = P_{in} - export$$

$$\frac{dW_2}{dt} = Q_{12} - Q_{23} - Q_2 - ET$$

$$\frac{dW_3}{dt} = (Q_{23}/100) - (Q_3/100) + (100(-\sum_{i=1}^{n} GW_{in_i} + \sum_{i=1}^{n} GW_{out_i})/UNDERC_{area})$$



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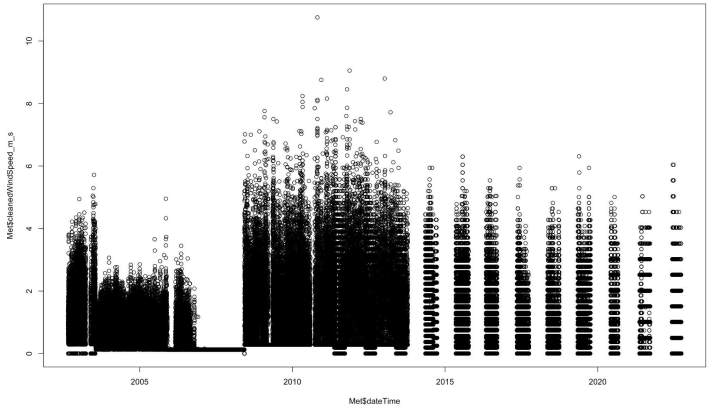
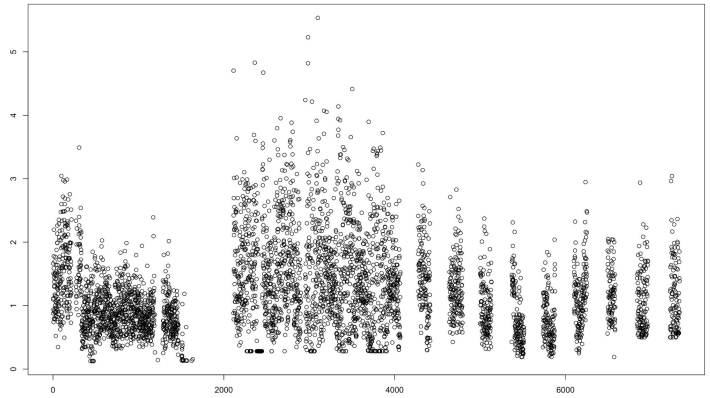
Parameters and Forcings

Data collected onsite by UNDERC:

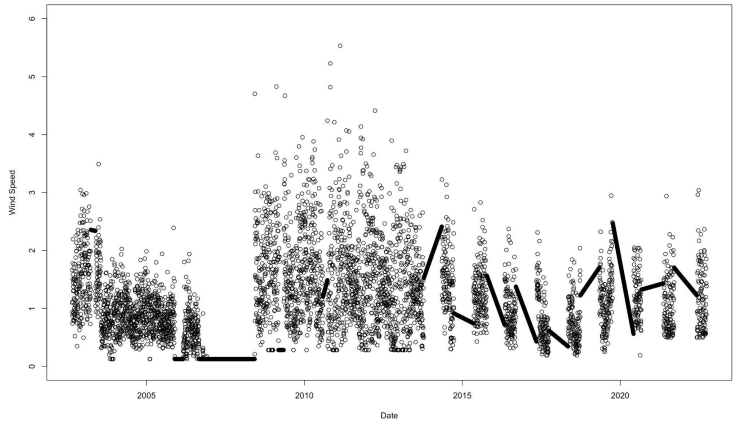
527,287 observations



4,425 usable daily averages



7,336 points



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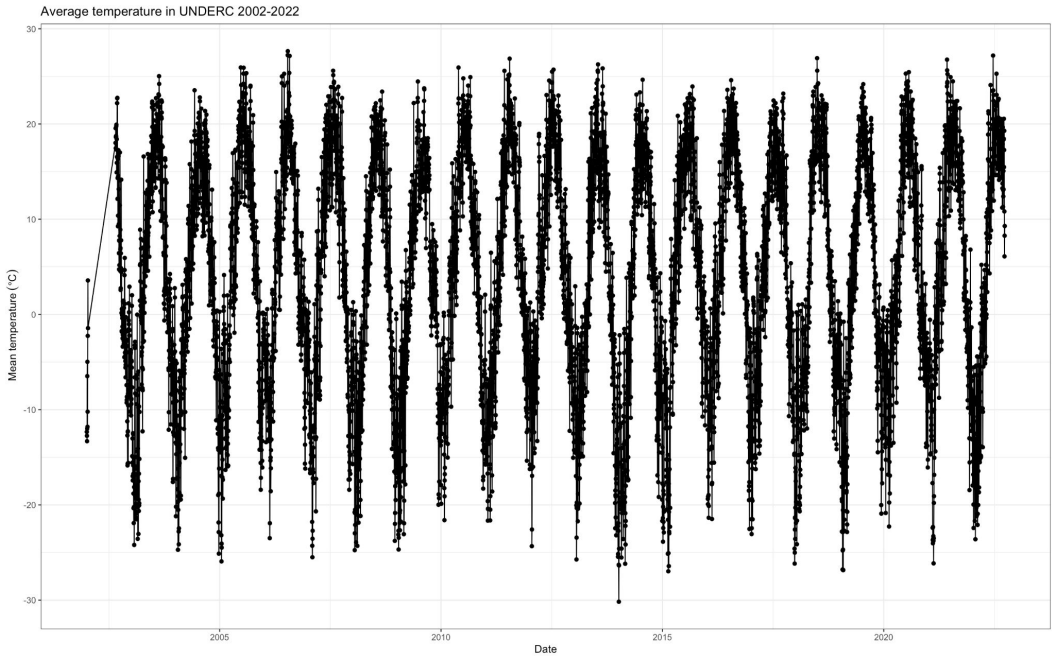
Parameters and Forcings

PRISM Data:

- Precipitation, air temperature, vpd, solar radiation are all available at different time steps
- First download data for the whole US then slice down to the data of a 4x4 km square in UNDERC to get our daily averages



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Parameters and Forcings

Constant Parameters:

- University of Colorado, Boulder - Water and Climate Research Group - VIC model parameters at 1/16°
- Constant across time and area
- hydraulic conductivity, residence time, max soil layer depths, ...

pName	value
bi	0.3
r	1
Ks	15.45
Bp1	0.55
Wmax1	10
Wmax2	50
Bp2	0.3
W20	10
Tstar	100
Tstar2	15
W30	40
pctInt/ev	0.06
Tstar3	16

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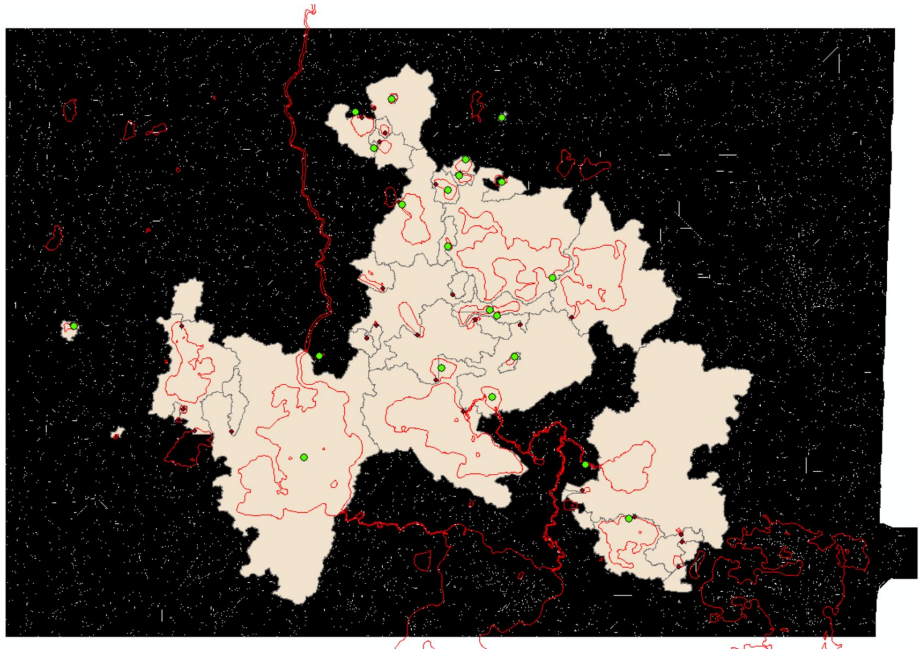
Parameters and Forcings



Lake-wise Data:

- Going through different data tables to get lake surface area, depth
- Using the DTM to get lake surface elevations
- Using ArcGIS for catchment area and perimeters

	lake_id	lake_surface_elevation	lake_depth	lake_surface_area	avg_depth	catchment_area	perimeter
1	TU	521.05	14.93	0.91	7.500000	104346.204	487
2	RB	517.88	6.09	4.62	3.044000	121707.522	884
3	PA	516.70	11.99	1.66	6.000000	35182.345	472
4	PE	516.68	18.00	2.52	9.000000	68399.556	599

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Putting Code Together

Equations:

- Reading ecology/hydrology papers on similar topics to piece together equations I need

$$E_{\text{mass}} = \frac{mR_n + \rho_a c_p (\delta e) g_a}{\lambda_v (m + \gamma)}$$

7 parameters needed additional equations to be calculated, at least 4 data inputs needed

m = Slope of the saturation vapor pressure curve (Pa K^{-1})
R_n = Net irradiance (W m^{-2})
ρ_a = density of air (kg m^{-3})
c_p = heat capacity of air (J kg^{-1} K^{-1})
δe = vapor pressure deficit (Pa)
g_a = momentum surface aerodynamic conductance (m s^{-1})
λ_v = latent heat of vaporization (J kg^{-1})
γ = psychrometric constant (Pa K^{-1})

$$E_{\text{PEN}} \approx 0.051(1-\alpha)R_S\sqrt{T+9.5} - 2.4\left(\frac{R_S}{R_A}\right)^2 + 0.052(T+20)\left(1-\frac{RH}{100}\right)(a_U - 0.38 + 0.54u).$$

2 parameters needed additional equations to be calculated, 3 data inputs needed

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Putting Code Together

Structure:

Script 1: Getting data

Reading in data, matching time steps, interpolating missing days

- Lake characteristics
- Atmospheric data

Script 2: Putting the ODE together

ODE function: calculate state variables for time range

calling forcings for given time step

- calculate rates of atmospheric fluxes
- calculate infiltration rate in the three soil layers
- ROC of volume

return a list of the state variables for each time step and plot

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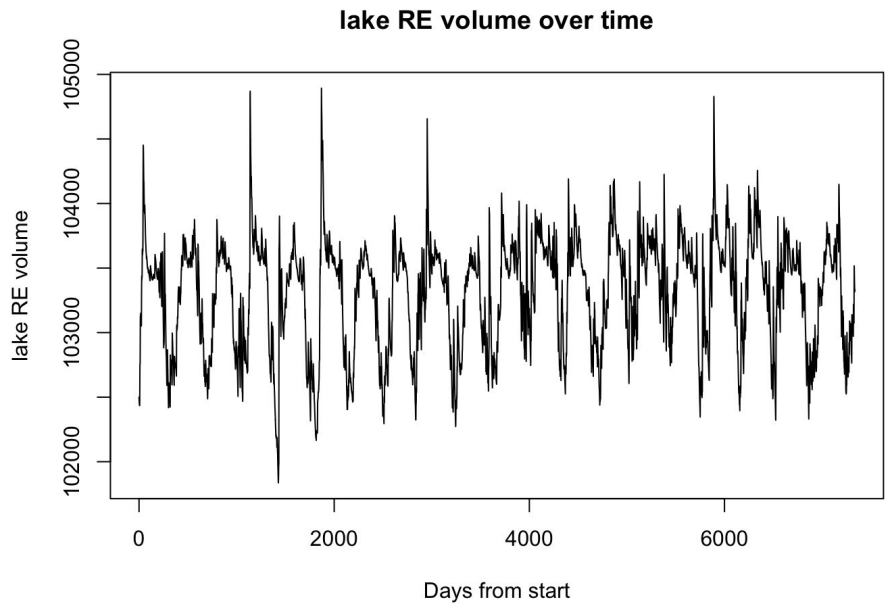
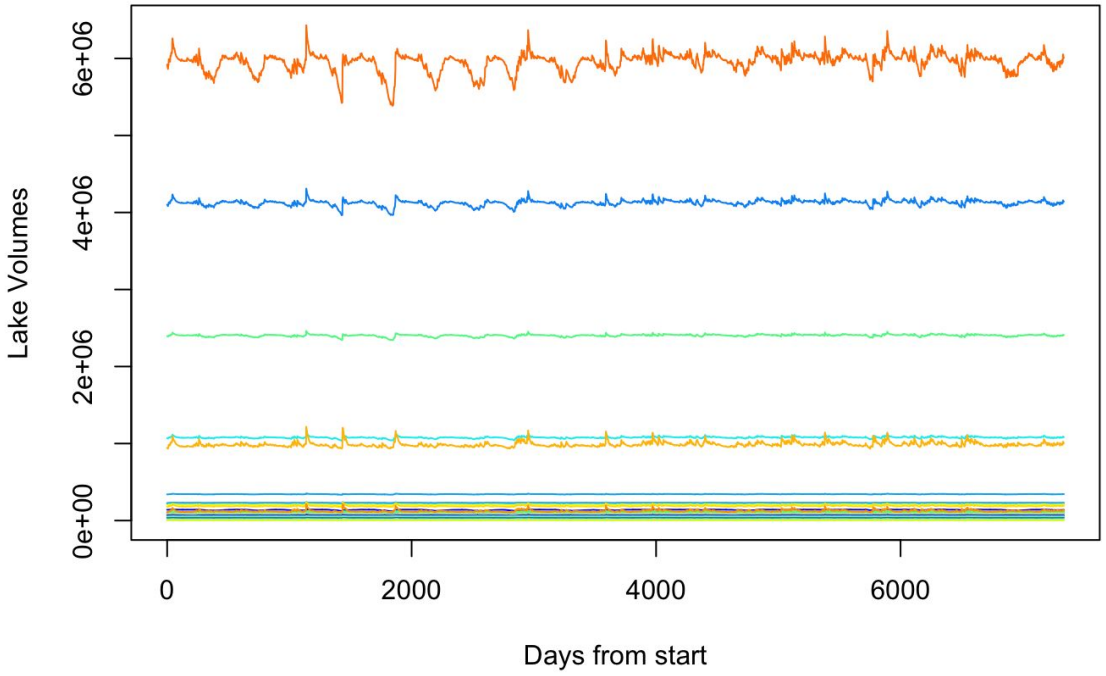
Results and testing

- compare to known lake volumes over time
- fitting constant parameters





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Whats next for the project

- Code will be available to UNDERC researchers
- Water flows can be used to see how chemicals travel through different lakes
- Soil layer depths and water table elevation affect the plants and trees, there is interest in how it changes over time

scan for access to github repository:

https://github.com/lschulz3/CICF-UNDERC-summer24

Acknowledgments and References

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Any Questions?